VF-E04 Heat Aging Test

Test Method for Heat Aging Test	Test method # VF- E04 REF #:
Date Issued: September 15, 2022 Revision #: 2	Reason for Revision: 1. Revise test procedure step 3 and 4 2. Revise reporting for material and shoes

PURPOSE AND SCOPE: To measure the tendency of the surface of a material to discolor as a result of exposure to oven heating.

REFERENCE: ASTM D 1148

EQUIPMENT AND MATERIAL:

1. Air-circulating oven, capable of maintaining 70+/-1°C
2. AATCC Chromatic Transference Scale/Grey Scale
3. Plastic, rubber, fabric, leather, synthetic leather, printed material /component, trims, and assembled shoes.

SAMPLE PREPARATION:

Condition the test samples at 23 ± 2°C and 55 ± 10% relative humidity for at least 24 hours prior to testing. Samples with water base cement should be conditioned for 48 hours after assembly unless the cement supplier has specified that 24 hours is already sufficient for conditioning period.

1. **For Finish footwear:** Test the whole footwear at least 1 foot
2. **For material:** Die or hand cut the test sample 6 cm long *3cm wide

REPETITIONS:

- Finished footwear: at least 1 foot (un-cut)
- For material: test two (2) samples per material.

TEST PROCEDURE:

1. Mask the first and last 2cm of the test sample with a piece of non-transparent masking tape this will leave the center 2cm unmasked. For assembled shoes, place one foot of shoes in an oven.
2. Place the test materials in an oven at 70+/-1°C for 3 days or 7 days.
   - Place shoes in an oven at 70.0+/-1.0°C for 3 days.
   - ***For color migration, testing should be for 7 days at 70 +/-1.0°C***
3. After removing the specimen/shoe from the oven, let it stabilize for 24 hours at 23± 2°C and 55±10% relative humidity.

4. Place the test specimen in light box, set for 'Daylight' (D65). Examine the change in color - compare the color of the darkest portion of the uncovered area of the test sample to the AATCC Chromatic Transference Scale/Grey Scale and rate to the nearest scale – rating.

Note: Chromatic Transference Scales / Grey Scale must be replaced at least every 12 months.

REPORT:

- **For all materials:**

  Rating 1-5 AATCC Chromatic Transference Scale or Grey Scale

The rating is reported to the nearest 0.5 unit based on the AATCC Chromatic Transference Scale or Grey Scale.

- **For assembled shoes:**

  - Any delamination except delamination which cause by material shrinkage
  - Color change, color migration, bleeding for all material, components, and shoes
  - Metal: No corrosion
  - Printed: Peel off, bubbles, blisters, etc.